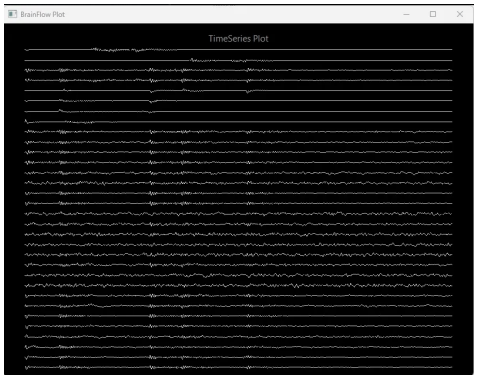
DATA INTERFACE & SOFTWARE GUIDE · SKU PIA24025P

Serial Protocol, SDKs & Firmware

IronBCI-32 appears on the computer as a USB serial port and streams one frame per sample as soon as it is powered. This guide covers the frame format, conversion to microvolts, BrainFlow, PiEEG Server, the example scripts, the open dataset and firmware updates.



921 600	107 B	~500/s	0.03725	ID 65
Baud · USB CDC-ACM	Frame, current firmware	Frames per second	µV per LSB	BrainFlow board ID

01 SOFTWARE OPTIONS

Option	Best for	Runs on	Install / setting
PiEEG Server	Dashboard, recording, LSL, OSC, WebSocket stream to any language	Windows, macOS, Linux	`pip install "pieeg-server[ironbci32]"`
BrainFlow	Research pipelines in Python, C++, Java, C#, R, Julia, MATLAB	Windows, macOS, Linux, Raspberry Pi	`BoardIds.IRONBCI_32_BOARD`
PiEEG Cloud	Fastest start, dashboards, spectra and topographic maps	Chrome or Edge (Web Serial)	None — cloud.pieeg.com
JavaScript SDK	Browser applications	Chrome or Edge	`connectSerial({ device: 'ironbci-32' })`
Example scripts	Plotting and saving with BrainFlow	Python 3.12, BrainFlow 5.20.1	Clone the repository (SDK/)

02 SERIAL DATA FORMAT

Port	USB CDC ACM (virtual COM port) on the Micro-USB “Data Stream” connector
Baud rate	921 600
Start	Streams as soon as the board is powered. No start command.
Frame rate	One frame per sample, about 500 per second

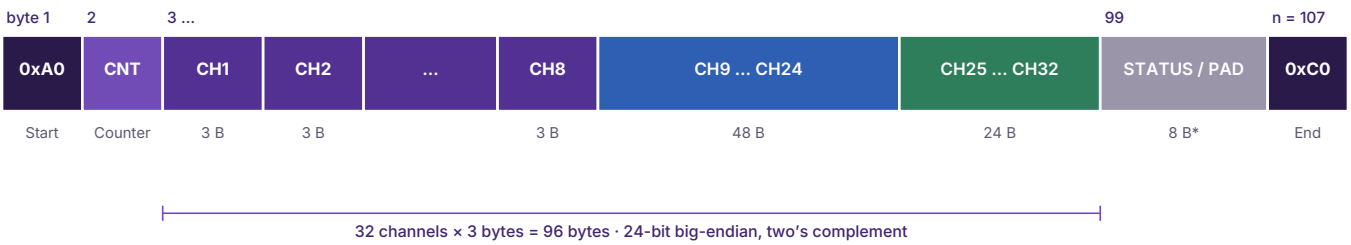


Figure 1 — Frame layout. * Status/padding length depends on the firmware; frames on current firmware are 107 bytes.

Bytes	Content
1	`0xA0` start marker
2	Sample counter (0–255, wraps) — use it to detect dropped frames

Bytes	Content
3–98	32 channels × 3 bytes, 24-bit big-endian two's complement, channel 1 first
99 … n−1	Status and padding bytes (length depends on the firmware)
n	`0xC0` end marker

NOTE Robust parsing.
Find the frame length from the distance between consecutive `0xC0 0xA0` pairs instead of assuming a fixed size. For production use, BrainFlow or PiEEG Server handle frame detection and errors for you.

03 CONVERTING TO MICROVOLTS

PiEEG Server and BrainFlow use a 2.5 V reference and a gain of 8:

$$\mu V = raw \times 2.5 / 8\,388\,607 / 8 \times 10^6$$

One count is about **0.03725 µV**. Raw values carry a large DC offset (tens of thousands of µV) from the electrode contact — that is normal. Apply a high-pass or band-pass filter before looking at EEG.

PYTHON
```
raw = (b[0] << 16) | (b[1] << 8) | b[2]
if raw & 0x800000:       # negative value
    raw -= 1 << 24
uV = raw * 2.5 / 8388607 / 8 * 1_000_000
```

04 MINIMAL READER (PYSERIAL)

PYTHON · READER FOR TESTING AND CUSTOM FIRMWARE WORK
```
import serial

PORT = "COM3"                       # or "/dev/ttyACM0"
FRAME = 107                         # check with the 0xC0 0xA0 method if unsure

with serial.Serial(PORT, 921600, timeout=1) as s:
    buf = bytearray()
    while True:
        buf += s.read(1024)
        i = buf.find(b"\xA0")
        while i >= 0 and len(buf) - i >= FRAME:
            f = buf[i:i + FRAME]
            if f[-1] == 0xC0:
                ch = []
                for c in range(32):
                    b = f[2 + 3*c: 5 + 3*c]
                    raw = (b[0] << 16) | (b[1] << 8) | b[2]
                    if raw & 0x800000:
                        raw -= 1 << 24
                    ch.append(raw * 2.5 / 8388607 / 8 * 1e6)
                print(f[1], [round(v, 1) for v in ch[:4]], "...")
                buf = buf[i + FRAME:]
            else:
                buf = buf[i + 1:]
            i = buf.find(b"\xA0")
```

05 BRAINFLOW

IronBCI-32 is a native BrainFlow board: `BoardIds.IRONBCI_32_BOARD` (ID **65**). The only setting is the serial port. Repository scripts were tested with BrainFlow 5.20.1 and Python 3.12.4.

PYTHON · PIP INSTALL BRAINFLOW

```python
import time
from brainflow.board_shim import BoardShim, BrainFlowInputParams, BoardIds

params = BrainFlowInputParams()
params.serial_port = "COM3"              # or "/dev/ttyACM0"

board = BoardShim(BoardIds.IRONBCI_32_BOARD, params)
board.prepare_session()
board.start_stream()
time.sleep(10)
data = board.get_board_data()            # rows = channels, columns = samples
board.stop_stream()
board.release_session()

eeg = BoardShim.get_eeg_channels(BoardIds.IRONBCI_32_BOARD)
print(data[eeg].shape)                   # (32, number of samples)
```

CAUTION Serial-port permission on Linux / macOS.

Add your user to the dialout group (`sudo usermod -aG dialout $USER`, then log in again) or run with sudo.

06 PIEEG SERVER

SHELL

```
pip install "pieeg-server[ironbci32]"          # Python 3.10+, installs pyserial
pieeg-server --device ironbci32 --serial-port COM3                      # stream + dashboard
pieeg-server --device ironbci32 --serial-port COM3 --filter             # 1-40 Hz band-pass
pieeg-server --device ironbci32 --serial-port COM3 --record session.csv   # record while streaming
pieeg-server --device ironbci32 --serial-port COM3 --lsl                # Lab Streaming Layer
pieeg-server record session.csv --device ironbci32 --serial-port COM3     # CSV only, no server
```

Address	What it is
`http://localhost:1617`	Browser dashboard
`ws://localhost:1616`	WebSocket stream: JSON frames with 32 channels in µV

07 EXAMPLE SCRIPTS (SDK FOLDER)

Script	What it does
`Real_time_visualisation/1.One_ch_plot.py`	Live plot of one channel
`Real_time_visualisation/2.All_channels.py`	Live plot of all 32 channels
`Save_data/1.Save_5_sec_1ch.py`	Record 5 seconds of one channel
`Save_data/2.All_ch.py`	Record 10 seconds of all 32 channels to Excel
`Not_real_time/Plot_data.py`	Plot a saved file with an 8–12 Hz filter and wavelet view

SHELL

```
pip install brainflow pyqtgraph PyQt5 pandas openpyxl scipy matplotlib PyWavelets
git clone https://github.com/pieeg-club/ironbci-32.git
cd ironbci-32/SDK/Real_time_visualisation
python 2.All_channels.py --serial-port COM3       # or /dev/ttyACM0
```

CAUTION Before running the save scripts.

The port `COM3` is written in the code — change it to yours. In `2.All_ch.py` fix the typo `IRONBBCI_32_BOARD` → `IRONBCI_32_BOARD`.

08 OPEN DATASET

`dataset/eeg_32ch_20s.xlsx` — raw EEG, 32 channels (`Ch1`–`Ch32`) in µV, 10 241 samples (about 20 s at 512 Hz), recorded with the PiEEG 32-channel wet cap. Channel numbers follow the 10-20 layout in IRB32-PIN-004. The plots below were computed from this file.

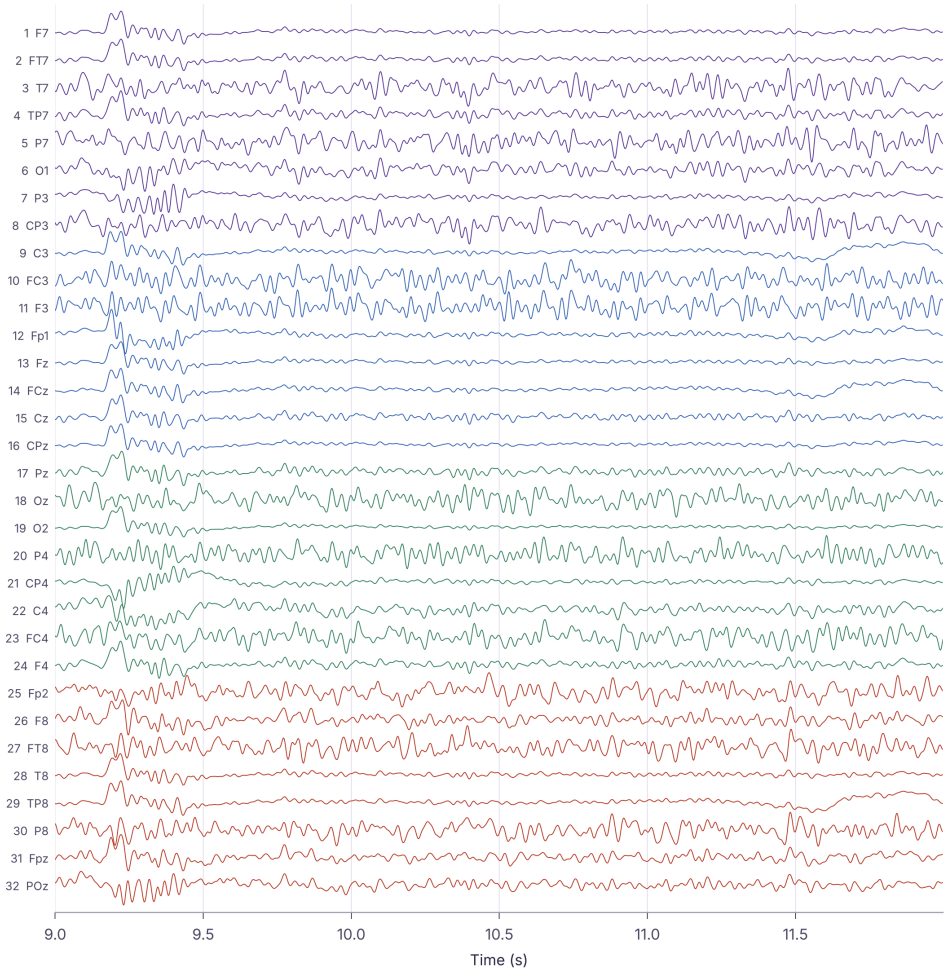


Figure 2 — 3-second excerpt, all 32 channels, 50 Hz notch + 1–40 Hz band-pass (4th-order Butterworth, zero-phase). Each trace is auto-scaled; colours follow the four headers.

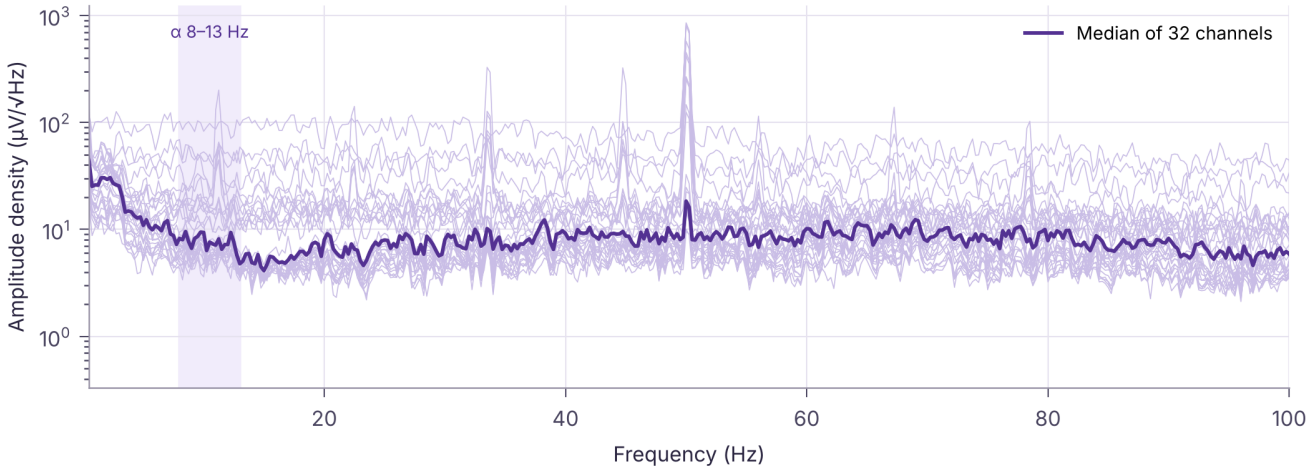


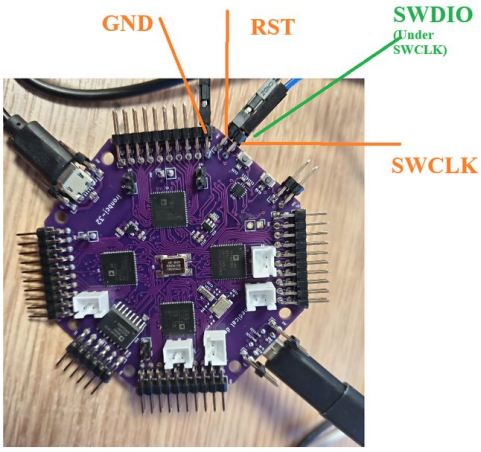
Figure 3 — Amplitude spectral density of the raw recording (Welch, 4 s segments). Thin lines: individual channels; bold: median of 32 channels. Shaded: alpha band.

09 FIRMWARE

IronBCI-32 ships with firmware installed. Update it only when asked to, or if the board stops creating a serial port with a known-good cable and power. Flash with **no electrodes on a person**.

Item	Details
ST-Link V2	Programmer with its 4 jumper wires
STM32CubeProgrammer	Free tool from STMicroelectronics (st.com/stm32cubeprog)
Firmware file	`hex/ironbci32.hex`
5 V USB battery	Powers the board while flashing

ST-Link	IronBCI-32
GND	GND
RST	RST
SWDIO	SWDIO
SWCLK	SWCLK



ST-Link wiring to the SWD pins.

1 Wire and power

Connect the ST-Link to the 4-pin SWD header as in the table. Plug the 5 V USB battery into **Power**, then plug the ST-Link into the computer.

2 Connect

Open STM32CubeProgrammer, choose **ST-LINK**, port **SWD**, and click **Connect**. If the ST-Link is not recognised, check its driver.

3 Download

Open `ironbci32.hex` and click **Download** to write it to the board.

4 Check

Disconnect the ST-Link, unplug power for a few seconds, power again and connect the data cable. The serial port should appear and stream as before.

Build from source: the STM32H743 project (Makefile, GCC Arm toolchain, OpenOCD scripts) is in `Framework/` (beta). Use the ready-made .hex unless you want to change the firmware.

10 RELATED DOCUMENTS

Doc. No.	Document	Contents
IRB32-DS-001	Product Datasheet	Features, specifications, block diagram, ordering
IRB32-MD-002	Mechanical & Dimensional Drawing	Outline, mounting holes, connector positions
IRB32-PCB-003	PCB Technical Information	Stack-up, layers, drill table, fabrication data
IRB32-PIN-004	Connector Pinout & Electrode Guide	Headers, pin order, REF/BIAS, 10-20 layout
IRB32-QSG-005	Quick Start & Safety Guide	Setup, battery-only operation, troubleshooting